Appendix for LLava-MS-PIT: Multi-Modal Schema-Guided Progressive Instruction Tuning for Multi-Modal Event Extraction

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Appendix A: Dataset Comparison and Error Analysis

A.1 ACE 2005 vs. M^2E^2(text): From Annotation to Evaluation

As mentioned in the subsection Further Analysis of Textual Event Schema Knowledge Injection, both experimental results and qualitative analysis of data characteristics indicate that the annotations in the M^2E^2 dataset are incomplete with respect to event information. To delve deeper into this phenomenon, we conduct a thorough case study by categorizing model prediction discrepancies into two main classes: predictions exceeding the gold annotations (**over-predictions**) and predictions missing from the gold annotations (**under-predictions**). Each class can be further subdivided into discrepancies arising from event predictions and those from argument predictions. Specifically, "over-predictions" include extra event types and additional arguments not annotated in gold data, whereas "under-predictions" encompass events and arguments that the model failed to predict.

	ACE 2005	**M^2E^2(text)**
Sentence Length Avg	165.3	176.3
Sentence Length Median	170	170.5
Event per Sentence	1.45	**1.19**
Args per Event	1.63	1.5
Over-Pred Event Rate	26%	**40.20%**
Over-Pred Args Rate	29.50%	**60.10%**
Under-Pred Event Rate	35.30%	14.10%
Under-Pred Args Rate	31.30%	9.80%

Table 1: Comparison of statistics between ACE 2005 and M2E2 datasets.

We perform a statistical comparison between the ACE 2005 and M^2E^2 test sets, with detailed results summarized in Table 1. ACE 2005, as a widely recognized and long-established annotated dataset, demonstrates relatively balanced metrics concerning over-prediction and under-prediction rates. In contrast, despite similar distributions in sentence length (average length: 165.3 for ACE 2005 vs. 176.3 for M^2E^2; median length: both about 170 tokens), M^2E^2 shows notable discrepancies in event density. Specifically, we observe a 18% lower average number of events per sentence and an 8% reduction in arguments per event compared to ACE 2005, suggesting annotation incompleteness in M^2E^2. Furthermore, the substantially elevated rates of over-prediction for both events (40.20%) and arguments (60.10%) on M^2E^2 further corroborate our hypothesis regarding the presence of unannotated event information in the dataset.

Over-Pred Events

Example 1:
Sentence: Laws seen as undermining LGBT rights in several Southern states have sparked protests and boycott threats from businesses and entertainers, as well as counterdemonstrations by conservatives.

Gold: {protests : Conflict-Demonstrate, Place : Southern states};

Pred: {protests : Conflict-Demonstrate, Place : states, Entity : businesses, Entity : entertainers}; **{counterdemonstrations : Conflict-Demonstrate, Place : states, Entity : conservatives};**

Example 2:
Sentence: A Second Night of Chicago Protests Over Police Shooting-Protesters took to the streets of Chicago for the second straight night Wednesday, one day after the city's police department released video of a black teenager being shot to death by a white officer.

Gold: {took : Conflict-Demonstrate, Entity : Protesters, Place : Chicago}; {death : Life-Die, Victim : teenager};

Pred: {Protests : Conflict-Demonstrate, Place : Chicago}; **{shot : Conflict-Attack, Attacker : officer, Place : Chicago, Target : teenager};** {death : Life-Die, Agent : officer, Place : Chicago, Victim : teenager};

Example 3:
Sentence: U.S. media reported that police arrested several people during the New York protests.

Gold: {arrested : Justice-Arrest-Jail, Agent : police, Place : New York};

Pred: {arrested : Justice-Arrest-Jail, Person : people, Agent : police, Place : New York}; **{protests : Conflict-Demonstrate, Place : New York};**

Example 4:
Sentence: ... Protest ban-Other recent demonstrations have been more tense than Monday's march, with scores of arrests following April protests against the pending transfer of two islands from Egypt to Saudi Arabia.

Gold: {arrests : Justice-Arrest-Jail};

Pred: {arrests : Justice-Arrest-Jail, Place : Egypt}; **{transfer : Movement-Transport, Artifact : islands, Agent : Egypt, Destination : Saudi Arabia}; {protests : Conflict-Demonstrate, Place : Egypt};**

Example 5:
Sentence: Algeria's government says at least 37 foreign nationals died after Islamists stormed the natural gas refinery at In Amenas last Wednesday.

Gold: {stormed : Conflict-Attack, Target : natural gas refinery, Place : In Amenas};

Pred: {died : Life-Die, Victim : nationals, Agent : Islamists, Place : refinery}; {stormed : Conflict-Attack, Place : refinery, Attacker : Islamists};

Example 6:
Sentence: Large-scale street demonstrations virtually vanished from Egyptian streets after security forces in August 2013 killed nearly 1,000 mainly Islamist protesters.

Gold: {killed : Life-Die, Place : streets};

Pred: {demonstrations : Conflict-Demonstrate, Place : streets}; {killed : Life-Die, Victim : protesters, Agent : forces, Place : streets};

Figure 1: Example of over-predicted events; over-predicted event information is shown in bold.

A.2 Analysis of Over-Predicted Instances

We present several representative examples illustrating instances of over-prediction from the M^2E^2. Figure 1 presents

Over-Pred Event Arguments

Example 1:
Sentence: There are several black men and women who have been killed in the hands of police.
Gold: {killed : Life-Die, Victim : women, Agent : police};
Pred: {killed : Life-Die, Agent : police, **Victim : men,** Victim : women};

Example 2:
Sentence: Egyptian security flank 52 men accused of sexual immorality as they arrive at a Cairo court November 14, 2001.
Gold: {arrive : Movement-Transport, Artifact : men};
Pred: {arrive : Movement-Transport, **Destination : court,** Artifact : men};

Example 3:
Sentence: The Istanbul governor said more than 160 foreign jihadists had been detained, along with 50 Turkish citizens during the two-week investigation.
Gold: {detained : Justice-Arrest-Jail};
Pred: {detained : Justice-Arrest-Jail, **Person : citizens, Person : jihadists, Place : Istanbul**};

Example 4:
Sentence: Pemberton, 21, an anti-tank missile operator from New Bedford, Massachusetts, has been detained at a compound guarded by Philippine and American security personnel, at the main military camp in metropolitan Manila, and not in a civilian jail.
Gold: {detained : Justice-Arrest-Jail, Place : Manila};
Pred: {detained : Justice-Arrest-Jail, **Person : Pemberton,** Place : compound};

Example 5:
Sentence: Workers prepare a tent camp in Khazer west of the Kurdish regional capital Erbil, Iraq, Oct. 11, 2016, for people expected to flee Mosul when the battles for Mosul begins.
Gold: {battles : Conflict-Attack, Place : Mosul}; {flee : Movement-Transport, Origin : Mosul};
Pred: {flee : Movement-Transport, **Artifact : people,** Origin : Mosul, **Destination : camp**}; {battles : Conflict-Attack, Place : Mosul};

Example 6:
Sentence: Malik, 29, and her American-born husband, Syed Rizwan Farook, 28, are now the principal focus of an enormous effort by U.S. authorities to discover what motivated or inspired the couple to plan and carry out the attacks, which primarily targeted Farook's co-workers in California's San Bernardino County, east of Los Angeles.
Gold: {attacks : Conflict-Attack, Place : California's San Bernardino County};
Pred: {attacks : Conflict-Attack, Attacker : couple, Place : San Bernardino County, **Target : co-workers**};

Example 7:
Sentence: Secretary Kerry will meet with senior representatives of Ukraine\u2019s new government, leaders of the Rada (Ukraine's parliament), and members of civil society.
Gold: {meet : Contact-Meet, Entity : Kerry};
Pred: {meet : Contact-Meet, Entity : representatives, Entity : leaders, Entity : members, Entity : Secretary Kerry};

Figure 2: Example of over-predicted event arguments; over-predicted event arguments is shown in bold.

representative cases where the model over-predicts events. Through manual examination of these cases, we observe that nearly all the over-predicted events are indeed grounded in the original sentences rather than generated from hallucinations by the model. Similarly, Figure 2 illustrates instances of arguments over-prediction. Further analysis confirms that these additionally predicted arguments logically serve as valid constituents for existing events. Taken together, these detailed case analyses, combined with the quantitative data characteristics discussed earlier, clearly demonstrate annotation incompleteness in the M^2E^2 dataset. This insight further explains the relatively low precision in textual event extraction observed in our primary experimental results.

Appendix B: imSitu-MME information

In this section, we present detailed information regarding the imSitu-MME dataset. Table 2 summarizes the comparison between the original event counts in the imSitu and the counts obtained after data cleaning (imSitu-MME). Table 3 provides the frequently used mappings between actions in imSitu and corresponding event types defined in ACE 2005.

Appendix C: Prompt for generating textual event schema

To systematically construct textual event schema instructions for multi-modal event extraction, we designed a series

Event Type	Origin data	Clean data
Transfer-Money	862	405
Die	1234	231
Arrest-Jail	2248	1311
Meet	2583	894
Attack	4887	1736
Demonstrate	1169	555
Phone-Write	935	934
Transport	5815	3202

Table 2: Comparison of event counts in imstiu before and after ACE mapping and cleaning.

Prompt for generating textual event schema 1:
You are an event-extraction analyst.

INPUT
Sentence: "<SENTENCE>"
Events: <TRIGGER1,TYPE1>; <TRIGGER2,TYPE2>; …

TASK
For each trigger–type pair:
• Quote the trigger exactly as it appears.
• Quote any additional word(s) or phrase(s) in the sentence that confirm the event.
• Provide a brief explanation (no length requirement, just concise) of how the evidence supports the event.

OUTPUT (exactly)
Event: <TYPE1>
Trigger: "<TRIGGER1>"
Evidence: "<phrase 1>", "<phrase 2>", …
Reasoning: <brief explanation>

Event: <TYPE2>
Trigger: …
Evidence: …
Reasoning: …

CONSTRAINTS
* Plain English only.
* Entire output ≤ 1000 characters (including spaces).
* Do not mention instructions, constraints, or add extra sections.

Prompt for generating textual event schema 2:
Based on the given sentence and event, explain the reasoning for why a specific event argument plays the specified role in the event. Clarify the concrete significance of this argument for the sentence and the event. Requirements: Keep your answer concise, logically clear, and within 1000 characters. Only include content directly relevant to the reasoning and explanation; do not output any unrelated text.

Input format:
Sentence: [sentence content]
Event Trigger AND Event Type : {event trigger1 : event type1}; {event trigger2 : event type2}; ...
Argument AND Role : {argument role11 : text11, argument role12 : text12, ...}; {argument role21 : text21, argument role22 : text22, ...}; ...

Output format:
Reasoning: [brief reasoning for why the argument plays this role in the event]
Significance: [explain the specific significance of the argument to the sentence and the event]

Figure 3: Prompt for generating textual event schema 1-2.

of curated prompts aligned with human cognitive processes, facilitating explicit reasoning and structural consistency in event extraction. Specifically, our textual event schema data generation comprises three distinct prompts:

(1) The first prompt explicitly instructs the model to identify event trigger and corresponding textual evidence, prompting concise reasoning to justify why these textual cues support this event, encouraging the model to learn clear trigger–evidence associations.

(2) The second prompt focuses on arguments identification, explicitly requiring the model to provide concise yet logically rigorous reasoning for assigning specific event arguments to particular event. By explicitly clarifying the significance of each argument to both the event and the sentence, this prompt enhances the model's structural understanding of events.

(3) The third prompt further advances the extraction reasoning process by implementing a step-by-step chain-of-

Event Type	Verb
Transfer-Money	gambling, pricking, paying, buying
Die	mourning, sleeping, crashing, falling, burying
Arrest-Jail	arresting, restraining, apprehending, dragging, detaining, handcuffing, catching, frisking, chasing
Meet	scolding, confronting, socializing, saluting, talking, leading, communicating, discussing, signing, shaking
Attack	colliding, burning, butting, flaming, deflecting, aiming, ejecting, punching, attacking, ramming, tackling, destroying, bulldozing, erupting, subduing, hitting, striking, tugging, shooting, launching, slapping
Demonstrate	cheering, parading, gathering, congregating, protesting
Phone-Write	phoning, telephoning, writing, calling
Transport	rafting, carrying, weighing, boating, towing, boarding, rowing, hauling, steering, landing, marching, wheeling, unpacking, floating, fetching, taxiing, lifting, piloting, loading, skidding, unloading, disembarking

Table 3: Mapping from verbs in the imStiu to event types in the ACE 2005.

Prompt for generating textual event schema 3:
You are an expert in event extraction. Given a sentence and its corresponding extracted events, your task is to generate a clear, structured, and concise step-by-step chain-of-thought reasoning process for identifying and extracting events according to the provided "event_mentions".

Strict Requirements:

Limit your total output to no more than 1500 characters.

The reasoning process must be brief, clear, and strictly focused on the extraction logic.

Only describe the necessary steps to identify triggers, event types, roles, arguments, and their relationships.

Avoid all redundant, repetitive, or unnecessary analysis or commentary.

Do not provide any extra explanations, disclaimers, background, or additional information.

Ensure the logic and conclusions strictly match the provided "event_mentions".

After your step-by-step reasoning, present the final extracted events exactly as shown in "event_mentions", using the following exact phrase and format, with no added text, headers, or formatting:

"Therefore, final event extraction results:\n[Answer]: "

Here is the sentence, along with its corresponding extracted events:

Figure 4: Prompt for generating textual event schema 3.

Prompt for generating visual event schema :
Given the image, generate a concise visual schema structured specifically for multimodal event extraction.

Only focus on critical objects that directly contribute to or indicate the occurrence of an event. For each critical object clearly list:
- object name
- visual attributes relevant to the event (e.g., color, size, state)
- actions performed by the object (if any)
- relations with other critical objects (as relation, object pairs)

Do NOT include irrelevant objects or details not related to potential events.

Output format:

Visual schema:
- **object:** object1
  attributes: attribute1, attribute2, ...
  actions: action1, action2, ...
  relations:
  - relation: relation type, object: related object
- **object:** object2
  attributes: attribute1, attribute2, ...
  actions: action1, action2, ...
  relations:
  - relation: relation type, object: related object
  ...

Scene: scene context

Figure 5: Prompt for generating visual event schema.

thought approach. It guides the model to clearly and systematically outline the cognitive steps necessary for identifying triggers, event types, arguments, and their semantic relationships, enhancing its capability in event structure reasoning.

The details of the prompts are provided in Figure 3 and Figure 4. By progressively structuring the prompts from basic trigger-evidence identification to detailed argument-role reasoning, and finally to rigorous chain-of-thought extraction logic, we effectively simulate hierarchical human cognitive processes in event extraction. This progressive prompting strategy ensures the injection of structural and semantic event schema knowledge into the model, significantly enhancing its event awareness and reasoning capabilities.

Appendix D: Prompt for generating visual event schema

The visual event schema prompt is explicitly designed to focus on critical visual objects that could potentially be related to event occurrences. The prompt clearly defines each critical object with its event-relevant visual attributes, actions, and relations with other key objects. This concise and structured representation ensures the prompt captures essential visual semantics, enabling cross-modal event concept alignment. The details of the prompts are provided in Figure 5.